What is still needed - 1 work, 1 claim

Six works were asked for last time; five are now closed. Carlson came whole and settled the sentence it was cited for, and the books that came with it carried Cattaneo, Vernotte and Danilovskaya. **One work is left**, and the table below it names the books that would do instead - every DOI on this page was resolved against a record on 2026-11-01.

If the 1912 volume cannot be found, do not spend another order on it. Born and Huang's *Dynamical Theory of Crystal Lattices* is Born's own treatise on lattice dynamics: the monatomic chain is solved there, which is exactly the relation the sentence states. If it credits the 1912 note the attribution survives; if it does not, the clean fix is to cite the book instead of the paper. Either way one book ends the list.

Still needed from you

#	Author(s) and year, then the title	What it unblocks	DOI	ok
1	**Born and von Karman 1912** *Über Schwingungen in Raumgittern* English: On vibrations in space lattices print only - Phys. Z. 13, 297-309; no publisher record and no scan found	The exact chain dispersion relation omega = 2 sqrt(kappa_s/m) times abs(sin(ka/2)).	*none*	-

Books that would do instead of the 1912 paper

#	For	Book, and why it would do	DOI or ISBN (verified)
1	this one CR181	**Born and Huang**, *Dynamical Theory of Crystal Lattices* 1954 (Oxford; Oxford University Press online reissue dated 1996) This is Born's own treatise on lattice dynamics and the standard place where the monatomic chain is solved, which is the relation the sentence states. If it prints omega = 2 sqrt(kappa/m) abs(sin(ka/2)) and credits the 1912 note, the attribution survives; if it does not credit the 1912 note, the clean fix is to cite this book instead of the 1912 paper. Either way one book settles the last open claim.	DOI 10.1093/oso/9780192670083.001.0001; ISBN 9780192670083 and 9781383002348; the chapter 'Lattice Vibrations' is DOI 10.1093/oso/9780192670083.003.0002
2	optional CR137 (optional)	**Nowacki**, *Thermoelasticity, second edition, revised and enlarged* 1986 (PWN, Warsaw, and Pergamon Press) Only if the author prefers to keep the Nowacki citation rather than drop it: the claim no longer blocks anything, because Carlson carries the sentence, but the book is the only way to read the Nowacki attribution itself. Beware the trap already recorded - the Nowacki book of 1975 is a different title, Dynamic Problems of Thermoelasticity (Noordhoff).	ISBN 978-0-08-024767-0; the chapters carry DOIs under 10.1016/b978-0-08-024767-0
3	optional CR092 and CR093 (optional, cross-check only)	**Straughan**, *Heat Waves* 2011 (Springer, New York) An independent modern book that sets out the Cattaneo-Vernotte law and its history; worth having only if a book-length confirmation of the relaxation-term story is wanted, since both claims are already carried by sources in hand. Not needed to unblock anything.	DOI 10.1007/978-1-4614-0493-4; ISBN 9781461404927 and 9781461404934

What closed this round, in one line each. CR092 - Cattaneo (1948) is credited with the relaxation term in a 2010 Oxford monograph, which says the equation becomes a telegraph, i.e. hyperbolic, equation. CR093 - Tzou calls the relaxed law the Cattaneo-Vernotte wave model and prints the 1958 note as C. R. 246, 3154-3155. CR136 - Carlson, sect. 24 gives the variational principles (pp. 338-342); sect. 13 (p. 319) and sect. 21 (p. 331) give the reciprocal theorems. CR168 - the 2010 monograph names problems "of Danilovskaya type" and prints her 1950 paper as Prikl. Mat. Mech. 14(3), in Russian.

One claim left the list without a book arriving. CR137 cites Nowacki for the same sentence as Carlson, and Carlson carries it in full. So the Nowacki reference is now a wording decision - drop it, or supply the book - and nothing blocks the claim. Note the trap recorded earlier: the Nowacki book of 1975 is *Dynamic Problems of Thermoelasticity*, a different title from *Thermoelasticity*.

Two things that were not usable. A one-page extract of Cattaneo's 1948 note was the front page of his 1966 Varenna course, with no argument on it - it did, however, confirm the venue, Atti del Seminario Matematico e Fisico dell'Universita di Modena, vol. 3. An engineering *Heat Conduction* text has no Cattaneo and no Vernotte in it at all. Neither was counted as evidence.

Outside the list: Polizzotto (2001), Int. J. Solids Struct. 38 - the held copy has no extractable text, so claim CR195 waits for a readable copy. And four claims (CR127, CR152, CR166, CR196) are recorded as held but no readable copy could be found; they need nothing from you.